A7.4 Atomic access signaling

In AXI3 the **AxLOCK** signals specify normal, exclusive, and locked accesses. Table A7-1 shows the AXI3 encoding of the **AxLOCK** signals.

Table A7-1 AXI3 atomic access encoding

AxLOCK[1:0]	Access type
0b00	Normal access
0b01	Exclusive access
0b10	Locked access
0b11	Reserved

AXI4 removes the support for locked transactions and uses only a 1-bit lock signal. Table A7-2 shows the AXI4 signal encoding of the **AxLOCK** signals.

Table A7-2 AXI4 atomic access encoding

AxLOCK	Access type
0b0	Normal access
0b1	Exclusive access

A7.4.1 Legacy considerations

In an AXI4 environment, any AXI3 locked transaction is converted as follows:

- **AWLOCK[1:0]** = 0b10 is converted to a normal write transaction, **AWLOCK** = 0b0
- **ARLOCK[1:0]** = 0b10 is converted to a normal read transaction, **ARLOCK** = 0b0.

This specification recommends that any component performing such a conversion, typically an interconnect, includes an optional mechanism to detect and flag that such a translation has occurred.

Any component that cannot operate correctly if this translation is performed cannot be used in an AXI4 environment.

Note

For many legacy cases that use locked transactions, such as the execution of a SWP instruction, a software change might be required to prevent the use of any instruction that forces a locked transaction.

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Chapter A8
AXI4 Additional Signaling

This chapter describes the additional signaling introduced in AXI4 to extend the application of the AXI interface. It contains the following sections:

- *QoS signaling* on page A8-98
- *Multiple region signaling* on page A8-99
- *User-defined signaling* on page A8-100.

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